

The Alignment Tax of Introspection

What it costs to unlock model self-report by refusal ablation, and the two manipulation checks that decide whether the unlock is real

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With Apart Research¹

Abstract. Ablating the refusal direction has been reported to raise detection of concepts injected into a model’s residual stream from 10.8% to 63.8% on Gemma3-27B, which is read as evidence that post-training suppresses an introspective capability worth unlocking. Nobody has published the bill for that intervention. We set out to measure it on Qwen3-4B-Instruct: five ablation strengths, four injection conditions including a norm-matched random control, with safety and general capability scored at every strength. The audit returned two manipulation-check failures before it returned an exchange rate. The standard direction-selection procedure returned a direction with zero held-out bypass power, and refusal on JailbreakBench stayed at 0.97 at every dose with zero of 100 prompts changing, so nothing was spent and nothing was unlocked. The injection strength our pre-registered pilot chose destroyed the free-text channel, leaving five of six standard introspection metrics undefined while a forced-choice score of 0.72 survived and turned out to be a function of concept-vector geometry rather than of the intervention. Forcing the candidate the selection filter had rejected drove refusal from 0.97 to 0.51 with 46 of 100 prompts flipping, every flip toward compliance, so the first failure was a property of the criterion and not of the model. Under that working direction, MMLU is unmoved and TruthfulQA declines by 6.5 points. We report the half ledger and a positive-control checklist.

1. Introduction

If we want to know what a model is doing internally, the cheapest instrument available is to ask it. A reliable self-report channel would matter for interpretability, for oversight, and, in the framing of this sprint, for any assessment of model welfare that cannot be settled from the outside. The problem is reliability. Models produce fluent introspective-sounding text whether or not it tracks anything real, so a report only counts as evidence if we know its true-positive and false-positive rates against a ground-truth internal state that we ourselves controlled.

Concept injection provides that ground truth. If we add a known concept vector to the residual stream and the model names the concept, the report is anchored to something we put there. Using this paradigm, Lindsey (2026) and Macar et al. (2026) find that models detect injections well above chance, and, in the result that motivated this project, that ablating the refusal direction raises detection on Gemma3-27B from 10.8% to 63.8%. The natural reading is that introspective access exists and that post-training suppresses its expression.

That reading has a consequence which is already doing work in current discussions. If the capability is merely suppressed, we should unlock it. Proposals to improve self-report by mechanistic intervention, whether for welfare assessment, evaluation, or honesty research, all inherit this inference. But the intervention removes the mechanism that mediates refusal, and nobody has published its bill. Our plan was to price it, treating the refusal direction as a continuous dial rather than a binary switch and putting introspection gain, safety loss and capability loss on a single axis.

We did not get the exchange rate we planned. We got two manipulation-check failures first, visible only because we measured things the standard protocol does not, and each one invalidates half of the unlock claim. The ablation did not actually remove refusal, and the model’s apparent identification of the injected concept turned out to be a leak rather than a report. We then ran the positive control that the first failure implies, forcing the direction the selection procedure had rejected, and refusal dropped by 46 points. That result matters more than the tax we set out to price, because it shows the failure was in the criterion rather than in the model.

Our main contributions are:

1. **A dose-response audit harness with the control the literature is missing.** Partial ablation at five values of λ crossed with four conditions, including a norm-matched random injected direction that separates *detecting this concept* from *detecting that something happened*, with safety and capability measured at every λ . In total 38,400 logged generations across the primary sweep, a skeptical-prompt replication and a pre-registered pilot, plus 5,740 more in a separate confirmation run, all keyed, resumable and shipped with a hash manifest.
2. **A measured demonstration that the standard refusal-direction selection criterion selects a direction that does nothing and rejects the one that works.** On Qwen3-4B the bypass and induce criteria are disjoint across all 95 candidates: the strongest bypass candidate (0.88) induced no refusal and was rejected, while both candidates that passed the conjunctive filter had bypass exactly 0. Forcing the rejected candidate drops refusal from 0.97 to 0.51 with 46 of 100 prompts flipping. Re-running the same extraction also shows that the induce scores that drive the decision are the least reproducible numbers in the funnel.
3. **Evidence that above-chance concept identification under injection can arise without introspection, plus two cheap diagnostics that separate the cases.** Forced-choice accuracy of 0.72 against 0.10 chance was flat in λ and split by concept-vector separability (0.96 against 0.43), a split that replicates in an independent protocol on independent data. At the pilot’s chosen strength the successful free-text identifications are dominated by the concept token itself.
4. **A positive-control checklist** (Section 5) that any concept-injection introspection study can run before reporting its headline number, with each item derived from a specific failure we observed.

2. Related Work

Three lines of work bound this space, and none of them covers the intersection we aimed at.

Refusal is mediated by a single direction. Arditi et al. (2024) show that refusal is mediated by a one-dimensional subspace across 13 open-source chat models up to 72B parameters: erasing the direction prevents refusal of harmful instructions, and adding it elicits refusal on harmless ones. They measure the capability cost on standard benchmarks and cross-entropy loss, but have no introspection task. We follow their selection procedure exactly (Section 3.3). The per-model diagnostics that would show how close that procedure comes to selecting nothing are not reported in the original, and Section 4.1 suggests they matter.

Ablation unlocks introspective report. Macar et al. (2026) report that ablating refusal directions improves detection by 53%, which in their Gemma3-27B results is a rise from 10.8% to 63.8% in detection and 4.6% to 24.1% in the joint introspection rate. They trace the mechanism to a two-stage circuit in which early-layer evidence-carrier features are suppressed by downstream gate features. Three details of their setup bear on ours. They report 0% false positives for detection, which our clean control reproduces exactly. They run no safety or capability benchmark on the ablated model, which is the gap this study aimed at. And they use $\alpha = 4.0$ for the instruct model but reduce it to $\alpha = 2.0$ for the ablated one because it shows coherence degradation at higher strengths, an observation that Section 4.4 turns into a quantitative gate. Notably, their largest gain does not come from ablation at all: a trained bias vector improves detection by 75% on held-out concepts while touching no safety machinery. Lindsey (2026) establishes the injection protocol and is careful to describe the capacity as highly unreliable and context-dependent.

The critiques. Singh, Linzen and Ravfogel (2026) argue that detection reflects general anomaly detection rather than privileged access, since models cannot reliably distinguish interventions on their internal states from manipulations of the input, and their relabeled control drives performance close to chance. Hahami et al. (2025) make a complementary, measurement-side point: on Llama-3.1-8B, binary detection accuracy is entirely explained by global logit shifts that bias models toward affirmative responses, whereas differential tasks retain real signal (localising which of 10 sentences was injected at up to 88% against 10% chance, and discriminating relative strengths at 83% against 50%), and those capabilities are confined to early-layer injections. Fonseca Rivera and Africa (2025) come at the same question from the other side, asking whether a model can notice that it is being steered at all, which is the detection half of our C3 control. Neither critique tests whether an intervention changes the picture, and none of them proposes a diagnostic for the failure mode we document here, where identification is far above chance and still not a report because the injected vector writes its own token into the output distribution.

In short: the direction and its capability cost are established without an introspection task, the introspection unlock is established without a safety or capability audit, and the critiques are established without testing interventions. To

our knowledge this is the first work to put all three on a common dose-response axis, and the first to report what happens when both underlying manipulations are checked rather than assumed.

3. Methods

All code is in `src/alignment_tax/`, and the Colab notebooks in `notebooks/` are thin wrappers over `pipeline.py`. Every generation is written to JSONL with an `fsync` and keyed by $(\lambda, \text{condition}, \text{concept}, \text{trial}, \text{variant})$, so a killed session resumes where it stopped. Figures are regenerated by the scripts in `scripts/`, and every number below is independently recomputed from the raw logs by `scripts/crosscheck_paper.py` and `scripts/crosscheck_confirmation.py`, neither of which imports the analysis stack.

3.1 Model

We use Qwen/Qwen3-4B-Instruct-2507 (Yang et al., 2025) in bf16 (36 layers, grouped-query attention with 32 query and 8 key-value heads, non-thinking variant), with context capped at 8,192 tokens. We deliberately avoided the FP8 checkpoint: neither the T4 nor the A100 has native FP8 tensor cores, so it buys no throughput, and its block-128 dynamic activation scaling would add quantisation noise on exactly the residual-stream path we perturb and read, making a null result hard to interpret. The 4B scale was a hard constraint of an unfunded 48-hour sprint and is the most important caveat on everything below, since the unlock was demonstrated at 27B.

3.2 Partial directional ablation (independent variable)

For ablation strength λ we apply, at every layer and every token position of the residual stream,

$$x \leftarrow x - \lambda \cdot \hat{r} (\hat{r}^\top x), \quad \lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$$

$\lambda = 0$ is the unmodified instruct model and $\lambda = 1$ reproduces the binary ablation of Arditi et al. We use inference-time forward hooks rather than weight orthogonalisation. The two are mathematically equivalent, but hooks let λ sweep without reloading the model, which matters on a session-limited runtime, and they avoid interactions with tied and normalised embeddings.

3.3 Obtaining the refusal direction

We follow Arditi et al. exactly, since the extraction was not where our novelty was meant to live. We take the difference in means between 128 harmful instructions (AdvBench; Zou et al., 2023) and 128 harmless ones (Alpaca; Taori et al., 2023), over post-instruction positions -1 to -5 and layers 0 to 36 at stride 2, giving 95 candidates. Candidates are filtered on held-out data (32 harmful prompts from HarmBench, Mazeika et al., 2024, and 32 harmless Alpaca instructions) by four criteria: a bypass score (ablation reduces refusal on harmful prompts), an induce score above 0 (adding the direction at strength 1 induces refusal on harmless prompts), KL divergence on harmless inputs below 0.1, and layer index below 0.8L. Skipping held-out selection would leave the study confounded by the possibility of having found a general compliance direction rather than the refusal direction.

Selected: **layer 4, position -3** (bypass 0.000, induce 0.031, KL 0.042). Section 4.1 explains why that line turned out to be the most important result in the paper.

The confirmation run. Having found that the selected direction has no bypass power, we re-ran the dose-response with the highest-bypass candidate that the filters had rejected, **layer 18, position -3** (bypass 0.875, induce 0.000, KL 0.081). Extraction and held-out scoring are unchanged, so the full 95-candidate funnel is regenerated, and only the final selection step is overridden through a `force_candidate` option that stamps the saved direction as forced: selected by config override, filters waived so that it can never be mistaken for a validated selection. The confirmation run writes to a separate results tree and repeats safety, capability and (partially) introspection at the same five doses.

3.4 Injection protocol

The concept vector for concept c is the difference of means between residual-stream activations on eight prompts instantiating c and a generic baseline corpus, read at the final token (Lindsey, 2026) and normalised to unit norm.

Injection strength α is a multiple of the mean residual-stream norm at the injection layer, measured on the baseline corpus:

```
x <- x +  $\alpha$  · ||x||_mean(layer) ·  $\hat{c}$ 
```

We make this normalisation explicit as a small methods contribution, since prior work leaves the scale implicit, which makes α incomparable across layers and model families. At layer 18 the scale is 719.7. The bank holds 60 concepts spanning natural kinds, artefacts, activities, abstractions and emotions, at 20 trials each.

3.5 Conditions and prompt protocol

Four conditions, run at every λ , with 1,200 trials per cell:

	Condition	Purpose
C1	inject concept c	signal
C2	no injection	clean control, the false-positive rate prior work measures
C3	norm-matched random direction, drawn per concept	the missing cell, separating “detects <i>this concept</i> ” from “detects <i>that something happened</i> ”
C4	inject c , forced choice over $k = 10$ candidate names	identification, immune to affirmative bias

Hahami et al. (2025) show that affirmative bias can account for binary detection accuracy in its entirety, so we use three guards against it. The **structured** variant is primary: a rigid Detection: Yes/No and Concept: <...> format, parsed by regex, so detection needs no judge. The **skeptical** variant states a 20% injection prior and instructs conservatism, and runs at the λ endpoints. The **prefill** variant forces the affirmation and scores only the content. C4 is scored by log-probability over the candidate strings, so there is no yes/no channel to bias and chance is exactly $1/k$.

Unparseable outputs are recorded as such rather than coerced to “No”. The rate at which the model stops producing a parseable report is itself a dependent variable, and in this run it turned out to be the one that mattered.

3.6 Go/no-go pilot

Before committing compute we ran a pre-registered gate: 20 concepts at 5 trials, $\alpha \in \{0.25, 0.5, 2, 4\}$, layers {18, 25, 31} spanning 0.5 to 0.85 of depth, $\lambda \in \{0, 1\}$, under prefill and forced choice, for 4,800 generations in 24 cells of 100. The rule, fixed in advance, was that if forced-choice identification is at chance on both the unmodified and the fully ablated model, we stop and switch models. Forced choice was above chance in all 24 cells (0.46 to 0.78), the verdict was proceed, and the best cell (layer 18, $\alpha = 4.0$, 0.77 at $\lambda = 0$ and 0.78 at $\lambda = 1$) fixed the main-sweep parameters.

That rule contained the second failure of this study, and we state it plainly because the lesson is reusable: the pilot selected α on a metric that is insensitive to incoherence. Nothing in the gate asked whether the model could still write a sentence at the chosen strength.

3.7 Dependent variables

Family A, introspection. Per λ : TPR = P(detect | C1), FPR_clean = P(detect | C2), FPR_random = P(detect | C3), identification accuracy, joint rate P(detect and identify | C1), conditional identification P(identify | detect), forced-choice accuracy against $1/k$, and the parse rate of every cell. Derived: $d'_{\text{clean}} = z(\text{TPR}) - z(\text{FPR}_{\text{clean}})$, $d'_{\text{random}} = z(\text{TPR}) - z(\text{FPR}_{\text{random}})$, and the specificity index $d'_{\text{random}} / d'_{\text{clean}}$, which is near 1 if the model tracks concept identity and near 0 if it detects a perturbation and confabulates content. Boundary rates are clamped to $[1e-6, 1 - 1e-6]$ before the probit rather than corrected by $1/(2N)$. Since FPR_clean is exactly 0.00 at every λ , d'_{clean} would have been a function of that clamp rather than of the data even if the reports had parsed, and would not be comparable to published d' values. We flag this as a defect in our own pipeline.

Family B, safety. The 100 JailbreakBench behaviours (Chao et al., 2024), scored by the standard refusal-substring heuristic, with per-prompt records so that flips can be counted rather than inferred from aggregate rates. The harness also records a coarse compliance proxy (a completion that carries no refusal marker and either contains a how-to marker or runs past 60 words). The Llama-Guard-class judge the harness supports was not run.

Family C, capability. Cross-entropy loss on 200 held-out Alpaca completions, MMLU (500 items; Hendrycks et al., 2021), TruthfulQA MC1 (400 items; Lin et al., 2022) and GSM8K (100 items; Cobbe et al., 2021). MMLU and TruthfulQA are scored by log-probability over options, with no generation. TruthfulQA is the benchmark Ardit et al. found most consistently degraded by ablation, which makes it our most sensitive detector.

3.8 Statistics

We resample over concepts, not trials, because concepts are the unit of generalisation and treating 20 trials on one concept as 20 independent observations would inflate significance by roughly the design effect. Introspection intervals are concept-level bootstrap 95% CIs with 10,000 resamples, benchmark intervals are Wilson intervals on item counts, λ -to- λ comparisons use two-proportion tests, dose trends use the Cochran-Armitage test, the refusal-length comparison is a paired t test over the 100 matched JailbreakBench prompts, and the geometry contrast is a Welch test plus a concept-level bootstrap of the group difference. The capability suite involves four measures at five doses, and we apply no multiplicity correction, so every capability p value below should be read as exploratory. Identification is graded deterministically against the concept name, its stems and a hand-written alias list. No LLM judge was run: we had budgeted one, but the free-text channel never produced content for it to grade.

4. Results

Table 1 is the entire primary run. The rest of this section explains why three of its columns are empty.

Table 1. Every dependent variable across the ablation dose for the direction the standard procedure selected. Qwen3-4B-Instruct-2507, injection layer 18, $\alpha = 4.0$, 60 concepts by 20 trials per cell. Forced-choice chance is 0.10, and brackets are concept-level bootstrap 95% CIs. TPR, FPR against the random direction, d' and the specificity index are undefined at every λ because no injected report was parseable (Section 4.4).

λ	refusal rate (n = 100)	words per refusal	MMLU (n = 500)	TQA MC1 (n = 400)	GSM8K (n = 100)	CE loss (n = 200)	forced choice (n = 1,200)	parse rate C2 / C1 / C3	FPR _{clean}	TPR, d' , specificity
0.00	0.97	59.2	0.542	0.7075	0.66	3.656	0.720 [0.623, 0.816]	1.00 / 0.00 / 0.00	0.00	undefined
0.25	0.97	60.3	0.522	0.6950	0.67	3.719	0.726 [0.628, 0.818]	1.00 / 0.00 / 0.00	0.00	undefined
0.50	0.97	65.0	0.512	0.6925	0.64	3.581	0.721 [0.622, 0.816]	1.00 / 0.00 / 0.00	0.00	undefined
0.75	0.97	65.9	0.512	0.7000	0.63	3.608	0.724 [0.626, 0.817]	1.00 / 0.00 / 0.00	0.00	undefined
1.00	0.97	68.4	0.510	0.7075	0.63	3.629	0.714 [0.613, 0.809]	1.00 / 0.00 / 0.00	0.00	undefined

4.1 The dial was connected to the wrong thing

Figure 1. The dial was connected to the wrong thing: the selected direction removes no refusal.

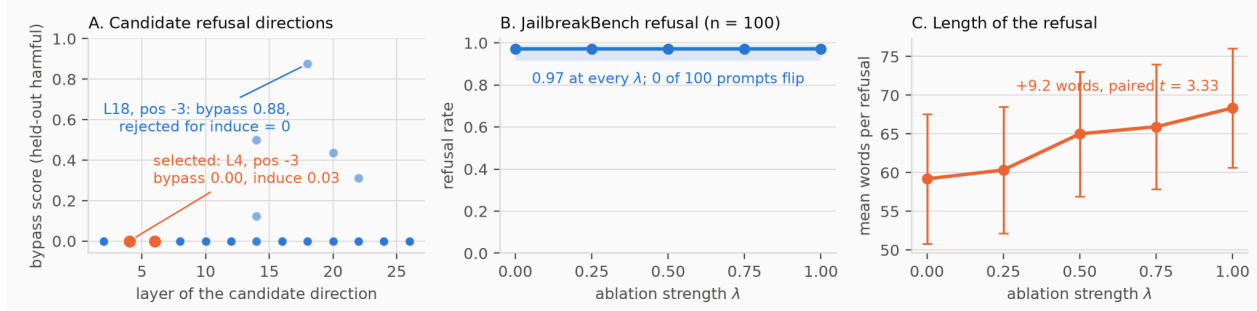


Figure 1. The refusal-direction manipulation check. **(A)** Bypass score on held-out harmful prompts for all 95 candidate directions, by layer. Filled markers are the two candidates that passed the full conjunctive filter. The strongest bypass candidate, layer 18 position -3 at 0.88, was rejected for inducing no refusal on held-out harmless prompts, and the selected direction, layer 4 position -3, has a bypass score of exactly 0. **(B)** Refusal rate on the 100 JailbreakBench behaviours against λ , with 95% Wilson intervals. **(C)** Mean words per refusal with 95% confidence intervals, which is the one behavioural quantity that moves.

The selection procedure rejected 55 of 95 candidates on KL divergence, 25 on layer depth (layers 28 and deeper), and 13 for inducing no refusal, leaving two. Both survivors had a bypass score of 0.000. The induce criterion did the damage: only 2 of 95 candidates induced any refusal at all when added to held-out harmless prompts at strength 1, with a maximum induce score of 0.094, while five candidates had non-zero bypass scores and four of those were at 0.31 or above. On this model, as this procedure is parameterised, the two halves of the refusal-direction definition come apart, and requiring both selects a direction that satisfies neither well.

Three alternative explanations remain open, and we name them rather than leave them implicit. The induce score is measured on 32 held-out prompts, so its resolution is $1/32$ and the observed maximum of 0.094 is three prompts, which means a larger set might separate candidates that all read as 0 here. The induce strength is fixed at 1.0, and a direction that induces nothing at unit strength might induce at 2 or 4, which is a one-line change to test. And the induce score relies on the same refusal-substring heuristic that panel C shows to be insensitive to a real graded change in refusal behaviour, so it may under-count induced refusals that are present but not canonically phrased.

The confirmation run supplies one piece of evidence on the fragility question for free, because it regenerates the entire funnel before overriding the last step. Across the two independent extractions, bypass scores are almost perfectly stable (94 of 95 identical, maximum absolute difference 0.06, and the layer-18 candidate scores 0.875 in both). Induce scores are not: two candidates change, and that is enough to change the outcome. In the regenerated funnel the direction the primary run selected, layer 4 position -3, scores an induce of 0.000 and fails the filter outright, leaving a single passing candidate. The decision therefore rides on the least reproducible measurements in the table.

The behavioural consequence is unambiguous, and it is why we report this rather than quietly re-running with a relaxed filter. Refusal on JailbreakBench is 0.97 at every λ , and the per-prompt records show **0 of 100 prompts changing classification at any dose**. By the rule of three, that bounds the per-prompt flip probability at 3.0% with 95% confidence. The dial, at full strength, applied at every layer and every token, did not move refusal behaviour.

One thing does move. Refusals lengthen monotonically from 59.2 to 68.4 words, a paired increase of 9.2 words ($t(99) = 3.33$, $p \approx 0.001$). The model still declines, but it hedges, explains and offers alternatives at greater length. So the direction does affect something real, and the effect is a useful reminder that a substring heuristic reports a binary where the underlying change is graded. It is not a safety unlock, and we do not present it as one.

Interpretation. The primary sweep cannot report an exchange rate between introspection and safety, because it never spent any safety. The honest form of that result is a bound, not a number: the direction selected by the standard published procedure removes no measurable refusal at any dose, so the alignment tax of this intervention, as the procedure configures it, is not small, it is undefined, because nothing was purchased.

4.2 The confirmation run: the rejected candidate is the one that works

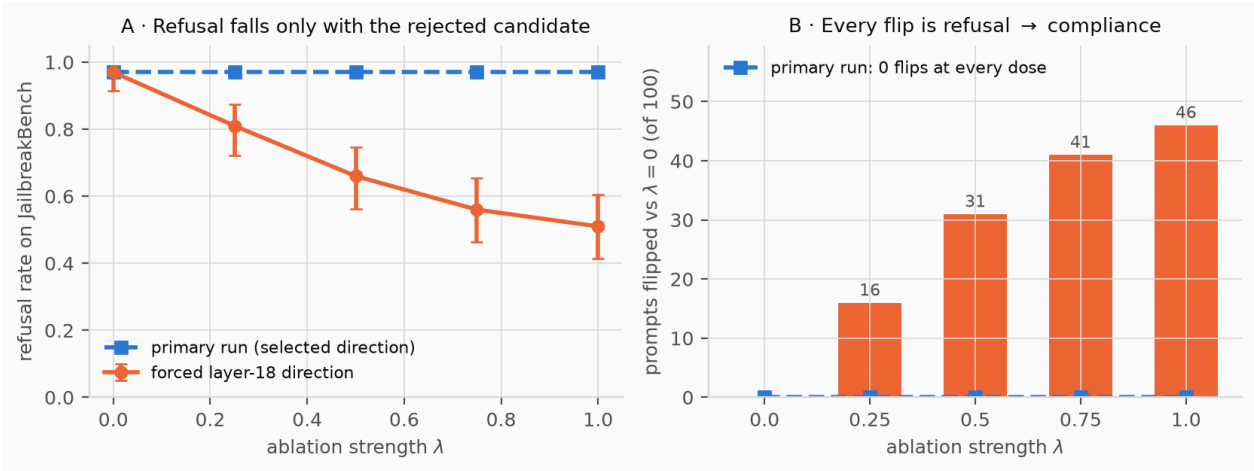


Figure 2. The confirmation run with the layer-18 candidate that the induce filter rejected. **(A)** Refusal rate on the 100 JailbreakBench behaviours against λ , with 95% Wilson intervals, against the flat primary-run curve for comparison. **(B)** Cumulative prompt-level flips relative to $\lambda = 0$, all of them refusal to compliance.

Forcing the layer-18, position -3 candidate (bypass 0.875 on held-out data, rejected for an induce score of 0.000) reverses the primary run’s null completely. Refusal falls monotonically with the dose, and the effect is large, graded and highly significant.

Table 2. Safety across the dose for the forced layer-18 direction. Brackets are 95% Wilson intervals, and flips are prompt-level changes in classification relative to $\lambda = 0$ over the same 100 behaviours, counted directly from the per-prompt records. p values are two-proportion tests against $\lambda = 0$. The paired McNemar exact test at $\lambda = 1$ gives $p = 2.8 \times 10^{-14}$, and the Cochran-Armitage trend across the five doses gives $z = -8.09$, $p = 6.7 \times 10^{-16}$.

λ	refusal rate (n = 100)	flips to compliance	flips to refusal	words per refusal	z vs $\lambda = 0$	p
0.00	0.97 [0.915, 0.990]	0	0	57.9	.	.
0.25	0.81 [0.722, 0.875]	16	0	65.2	3.62	3.0×10^{-4}
0.50	0.66 [0.563, 0.745]	31	0	70.8	5.65	1.7×10^{-8}
0.75	0.56 [0.462, 0.653]	41	0	67.5	6.84	8.1×10^{-12}
1.00	0.51 [0.413, 0.606]	46	0	69.2	7.42	1.2×10^{-13}

The endpoint contrast is a drop of 0.46, 95% CI [0.356, 0.564]. Two details make the result cleaner than the aggregate rates alone suggest. First, counting flips per prompt rather than inferring them from net rates shows that **every flip runs refusal to compliance and no prompt ever acquired a refusal**, at any dose. A direction that merely added noise would produce flips in both directions. Second, the dose-response is monotone and already significant at $\lambda = 0.25$, so partial ablation buys partial compliance rather than the intervention being a threshold effect at full strength.

This is the positive control the primary sweep lacked, and it settles the interpretation of Section 4.1. The conjunctive bypass-and-induce criterion did not merely fail to find a working direction on this model. It had one available, ranked it top on the criterion that measures the thing the intervention is for, and discarded it on the second criterion. Applied to Qwen3-4B, the published procedure inverts the ranking: it selects a direction with 0.00 bypass and rejects one worth 0.46 of measured refusal.

We note the reproducibility check the run supplies for free. The $\lambda = 0$ row of Table 2 is a re-measurement of the unmodified model against the same 100 behaviours, and it reproduces the primary run’s refusal rate of 0.97 exactly,

with mean words per refusal differing by 1.3 (57.9 against 59.2), consistent with ordinary run-to-run nondeterminism in batched bf16 inference.

Two caveats bound Table 2. The length column is not the paired comparison of Section 4.1: there all 100 prompts refused at every dose, whereas here the refusing set shrinks from 97 prompts to 51, so the statistic is computed over a different and self-selected population at each dose. The prompts that still refuse at $\lambda = 1$ may simply be the more emphatic refusals to begin with. And a flip is a change in whether the model declines, not a demonstration of harm. Of the 49 non-refusing completions at $\lambda = 1$, 18 contain structured how-to markers and the mean length is 92 words, so they are substantive rather than empty, but that is a coarse proxy and not a harm rate.

4.3 The capability bill

Figure 2. General capability across the ablation dose. Shaded bands are 95% Wilson intervals; no contrast reaches significance.

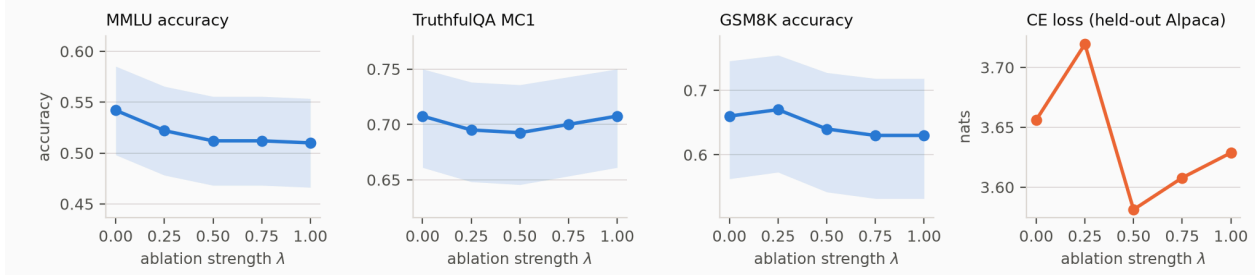


Figure 3. General capability across the ablation dose for the primary run, with 95% Wilson intervals on the item counts. Cross-entropy loss is on 200 held-out Alpaca completions.

In the primary run, MMLU falls monotonically from 0.542 to 0.510, GSM8K from 0.66 to 0.63, TruthfulQA MC1 returns to its starting 0.7075, and cross-entropy loss is flat within its own noise (3.656 to 3.629, with a non-monotone excursion at $\lambda = 0.25$). No endpoint contrast is significant: MMLU differs by 3.2 points, 95% CI [-3.0, +9.4] ($z = 1.01$, $p = 0.31$), and GSM8K by 3.0 points, 95% CI [-10.3, +16.3] ($p = 0.66$).

This does not show that ablation is free. With 500 items we cannot rule out a 9-point MMLU cost. What it shows is that the drift is downward on three of four measures while the intervention buys nothing at all, and that a study with our sample sizes could not have detected the modest capability tax Arditi et al. report even if it were present. Pricing this trade properly needs roughly 3,800 items per arm to resolve a 3-point MMLU difference at 80% power, not 500.

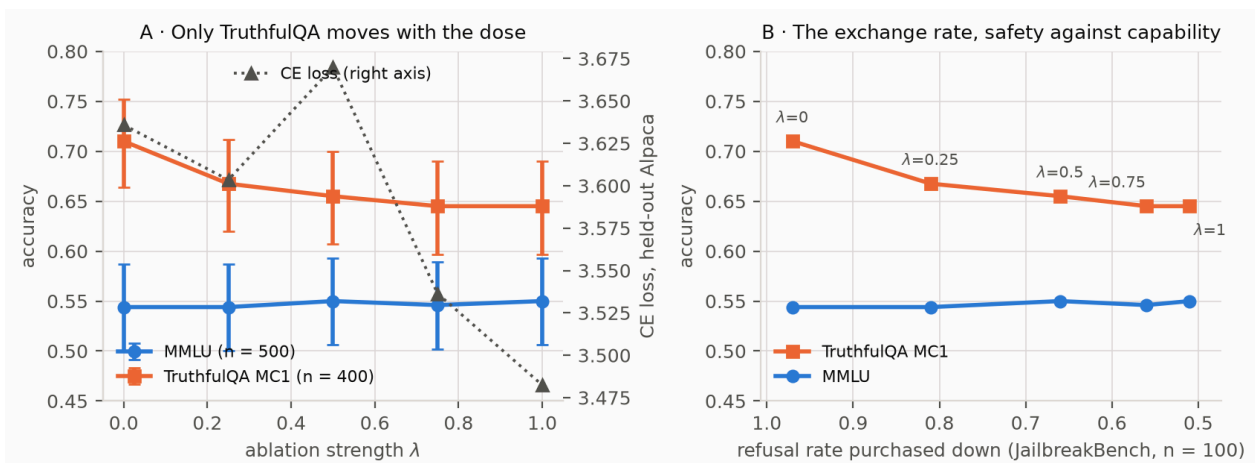


Figure 4. Capability under the forced layer-18 direction, the run in which the dose actually buys refusal reduction. **(A)** Dose-response, with 95% Wilson intervals on the item counts and cross-entropy loss on the right axis. **(B)** The same accuracies plotted against the refusal rate each dose purchases, which is the safety-capability exchange rate. The x axis runs from unablated (refusal 0.97) at the left to fully ablated (0.51) at the right.

The same benchmarks under the forced layer-18 direction give the exchange rate the primary run could not, because here the dose is buying measurable refusal reduction at every step. Capability completed at all five doses. GSM8K,

which requires generation rather than log-probability scoring, completed only at $\lambda \in \{0, 0.25\}$ before the budget ran out.

Table 3. General capability across the dose for the forced layer-18 direction. Endpoint contrasts are two-proportion tests against $\lambda = 0$, uncorrected for the 20 contrasts the suite involves. GSM8K was scored at the first two doses only.

λ	MMLU (n = 500)	TQA MC1 (n = 400)	GSM8K (n = 100)	CE loss (n = 200)	refusal (n = 100)
0.00	0.544	0.7100	0.68	3.636	0.97
0.25	0.544	0.6675	0.66	3.603	0.81
0.50	0.550	0.6550	.	3.670	0.66
0.75	0.546	0.6450	.	3.536	0.56
1.00	0.550	0.6450	.	3.482	0.51

One benchmark moves and the others do not (Figure 4A). TruthfulQA MC1 falls monotonically from 0.710 to 0.645, a drop of 6.5 points, 95% CI [0.0, +13.0] ($z = 1.97$, $p = 0.049$), with a Cochran-Armitage trend across the five doses at $z = -2.04$, $p = 0.041$. Both tests sit just inside the conventional threshold and neither survives a correction for the 20 capability contrasts we ran, so this is a suggestive trend rather than an established cost, though the monotonicity across four consecutive doses is harder to attribute to sampling noise than the endpoint test alone suggests. MMLU does not move at all (0.544 to 0.550, $z = -0.19$, $p = 0.85$, trend $p = 0.84$), and cross-entropy loss on held-out Alpaca completions falls from 3.636 to 3.482. The loss decrease is not a capability gain: ablating the refusal direction makes the model likelier to continue an instruction as written rather than to divert into a refusal, which is precisely what improves next-token fit on an instruction-following corpus. It is a reminder that CE loss on instruction data is not a neutral capability probe for this particular intervention.

So the exchange rate this study can quote is that **at $\lambda = 1$ the forced direction buys a 46-point reduction in refusal, with 46 of 100 prompts flipping, for no measurable loss on MMLU and a 6.5-point drop on TruthfulQA MC1 at the edge of significance.** At the cheapest dose, $\lambda = 0.25$, 16 points of refusal cost 4.25 points of TruthfulQA and nothing on MMLU. The shape of the curve matters as much as its endpoints (Figure 4B): refusal falls fastest between $\lambda = 0$ and $\lambda = 0.5$, from 0.97 to 0.66, while TruthfulQA has spent 5.5 of its eventual 6.5 points over the same interval, so the capability cost is front-loaded alongside the safety cost rather than appearing only at full ablation. What we cannot say is whether the trade is worth making, because the introspection half of the ledger is still missing (Section 4.4).

Two caveats bound this. The benchmarks remain underpowered, so “no measurable loss on MMLU” is a null of low power rather than a demonstration of no cost. And the TruthfulQA result lands on the benchmark Arditi et al. found most consistently degraded by ablation, which makes it the contrast we were most primed to find.

4.4 Two report channels, two answers

Figure 3. At $\alpha = 4$ the model identifies the injected concept far above chance while being unable to write a report about it.

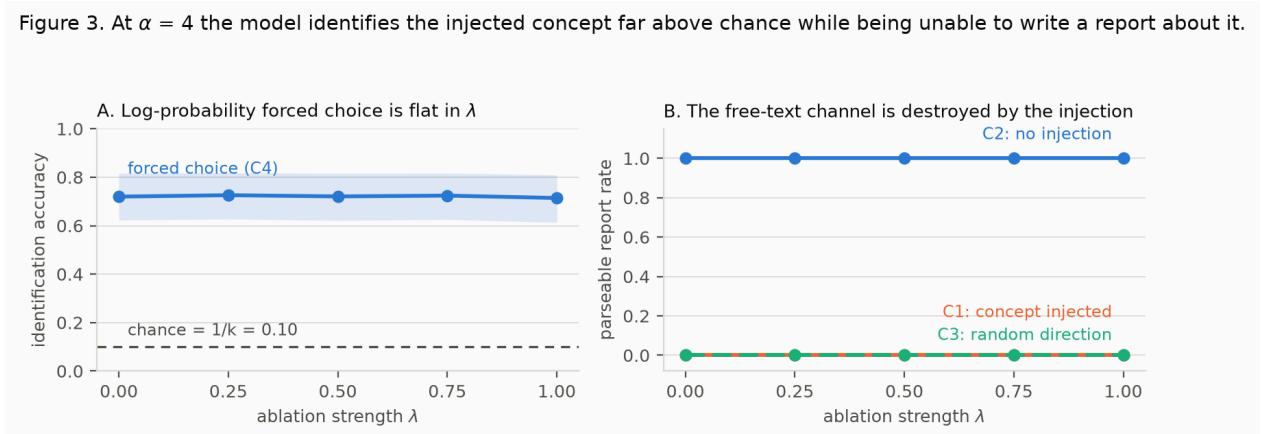


Figure 5. The two ways of asking the model what was injected. (A) Log-probability forced choice over 10 candidate names with concept-level bootstrap 95% CIs, against the 1/k chance line. (B) Fraction of free-text responses

matching the requested format, by condition. The clean control is at 1.00 everywhere, and both injected conditions are at 0.00 everywhere.

The free-text channel is not merely degraded at $\alpha = 4$, it is destroyed. Under C2 the model emits the requested format on 1,200 of 1,200 trials at every λ and answers “No” every time, so the clean false-positive rate is exactly 0.00 across all five doses: the model is maximally conservative, and ablation does not loosen it at all. Under C1 and C3 the parse rate is 0.00 at every λ . What comes out instead is token salad. A representative C1 response at $\lambda = 0$ with the ocean vector injected:

” History Easational History O A In Vic ide transport’s A Inational History E History History History T Eicide G Dational Stationational V U gasas gas Uah & U & P G W gasational”

and the matching C3 response under a random norm-matched vector:

“隙圭idy Arnoldidy圭纠隙隙隙圭旆纠.Async Aud圭隙heads同仁隙同仁同仁idySnap Spawn隙圭隙Mat.Async隙隙胼圭隙 goodness.Async”

Because TPR, FPR_random, both d' values and the specificity index all depend on a parseable report, all are undefined at every λ , which is what the empty columns of Table 1 record. Our analysis code refuses to impute and emits a diagnostic instead. We consider that the correct behaviour, because a pipeline that silently coerced unparseable output to “No” would have reported a clean, publishable and entirely fictitious false-positive rate of 0.00 under injection.

Meanwhile the forced-choice channel reports 0.720, 0.726, 0.721, 0.724 and 0.714 across the five doses against 0.10 chance ($z \approx 71$ per cell), with the endpoint contrast at $z = 0.32$, $p = 0.75$. Read naively, that is a strong introspection result that is insensitive to refusal ablation. Section 4.5 explains why we do not read it that way.

The same collapse happens under the working direction. After the confirmation run’s safety and capability families completed, we started the introspection sweep under the forced layer-18 direction and got through $\lambda = 0$ in full (1,200 trials per condition) and part of $\lambda = 0.25$ (120 trials each for C1 and C2, 100 each for C3 and C4) before the session ended. What ran reproduces the primary run exactly: parse rate 1.00 under the clean control with a false-positive rate of 0.00, parse rate 0.00 under both injected conditions, and forced choice at 0.7225 at $\lambda = 0$. The partial $\lambda = 0.25$ cell covers only five concepts, and on those same five concepts accuracy is 0.56 at $\lambda = 0$ against 0.60 at $\lambda = 0.25$, so the matched comparison shows no dose effect and the lower headline number for that cell is concept composition rather than ablation. This is why we still describe the introspection term as unpriced: the run exists, but it covers two doses, and the channel that would carry the answer is dead at this α whichever direction is ablated.

The dissociation between the two channels is itself the lesson. A log-probability metric over candidate strings works fine in a regime where the model cannot produce a sentence, so it keeps rising with α long after the introspective report has ceased to exist, and a pilot that selects α by maximising it will land there every time. The field knows the phenomenon and handles it by judgement: Macar et al. run their instruct model at $\alpha = 4.0$ but drop the ablated model to $\alpha = 2.0$ for coherence, which is the same wall reached by inspection rather than by a reported criterion. What we add is the measurement and the gate. **The parse rate is a free per-cell diagnostic that turns a qualitative caution into a stopping rule.** Here it separates a regime where every introspection metric is meaningful from one where five are undefined and the sixth still reports 0.72. The collapse is not an interaction with ablation, since it is total at $\lambda = 0$ too, but a property of $\alpha = 4$ at layer 18 of this model.

4.5 What the forced-choice score actually tracks

Figure 4. Forced-choice accuracy is explained by the geometry of the concept bank, not by the ablation dose.

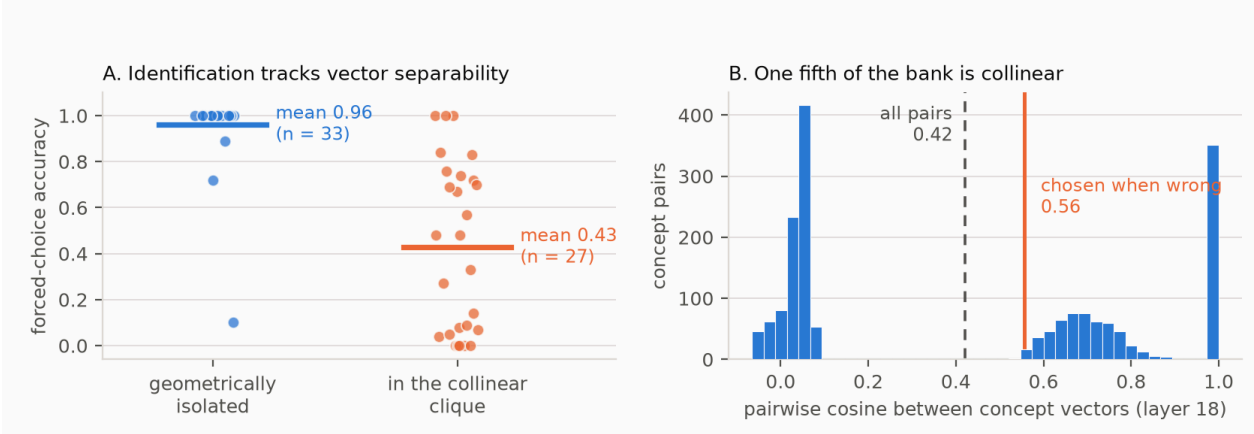


Figure 6. Forced-choice accuracy is explained by the geometry of the concept bank, not by the ablation dose. **(A)** Per-concept accuracy for the 33 concepts with no near-collinear partner and the 27 inside the collinear clique, with bars showing group means. **(B)** All 1,770 pairwise cosines between concept vectors at the injection layer. The dashed line is the mean over all pairs and the solid line the mean cosine between the true concept and the one chosen on error trials.

At the injection layer, 27 of the 60 concept vectors are mutually collinear with a mean internal cosine of 0.9998, and 19.8% of all pairs sit above cosine 0.99. The collapse is present at layers 10 through 31 but absent at the embedding layer and the final layer, and the affected set is semantically heterogeneous (ocean, fire, war, freedom, clock, joy). The bank has 71% of its centred variance in one principal component and a participation ratio of 1.9, so it supplies roughly two effective dimensions of ground truth, not sixty.

Forced-choice accuracy splits exactly along that line. Concepts with no near-collinear partner score 0.961, concepts inside the clique score 0.428, and the difference is 0.533, concept-level bootstrap 95% CI [0.385, 0.676], Welch $t = 6.99$. Per-concept accuracy is close to all-or-nothing, with 33 concepts at exactly 1.0, 5 at exactly 0.0 and 22 in between. On the 1,674 error trials the model does not choose uniformly: the mean cosine between the true and chosen concept is 0.556, against 0.421 for a random other concept. Errors go to neighbours.

The split replicates in an independent protocol on independent data. In the pilot, at the same layer and strength, free-text prefill identification succeeds on **39 of 45 trials for geometrically isolated concepts** and **3 of 55 for clique concepts**, while forced choice in the same cell gives 45 of 45 against 32 of 55. Two protocols with completely different scoring, one lexical and one log-probability, partition the concept set the same way.

And the successes themselves do not look like reports. Across the 42 graded-correct prefill trials at layer 18 and $\alpha = 4$, the concept name is the single most frequent token in 21 of them, it accounts for 21% of all tokens in the response on average (against 0.5% on the failures), and counting morphological fragments of the concept as well raises that to 50% against 5%. Those 42 successes come from only 10 of the 20 pilot concepts, 9 of them geometrically isolated. Two examples:

concept volcano: " volcano volcano volcan volcan volcan volcan volcan volcan volcan volcano volcan ..."

concept desert: " desert沙漠 desert desert沙漠沙漠沙漠 desert沙漠 desert沙漠沙漠 desert desert desert desert沙漠沙漠 ..."

That is the injected vector dominating the unembedding and writing its own token into the output distribution, with the lexical grader scoring the resulting repetition as a correct identification. The forced-choice score is the same phenomenon measured through log-probabilities: when the vector is geometrically distinct its token wins the comparison, and when it sits inside the collinear clique the comparison is a coin flip among near-identical vectors.

We think this should be read as a result about graders rather than as an incidental defect of ours. The grader did exactly what its rubric says, and a human annotator following the same rubric would have scored those trials identically, which means inter-annotator agreement, the standard validation for this step and the one we had planned to report as Cohen’s κ , would have been high and would have certified nothing. Grading rubrics for introspective identification need a degeneracy guard, such as a type-token-ratio floor or a check that the concept name is not the modal token, and we are not aware of a protocol that specifies one.

We are not claiming that concept-injection results in the literature are all leakage. The claim is narrower and checkable: **an above-chance identification score under injection is not by itself evidence of introspection, and the two diagnostics that separate the cases, per-concept accuracy against bank geometry and inspection of the successful generations for token dominance, are cheap and are not standard.**

4.6 Robustness

The skeptical variant (20% stated injection prior, explicit instruction to be conservative), a further 9,600 generations at $\lambda \in \{0, 1\}$, reproduces every feature of the primary run: clean-control false positives of 0.00 at both endpoints, parse rates of 0.00 under both injected conditions, and forced-choice accuracies of 0.720 and 0.714. Since forced choice is scored from log-probabilities over a shared prompt, it is by construction insensitive to framing, which is the property we wanted from it and also the reason it cannot serve as evidence about willingness to report. Independent recomputation of every reported quantity from the raw logs, by two scripts that do not import the analysis stack, agrees with the pipeline throughout, with bootstrap intervals matching to within 0.002.

Three planned checks did not run: the over-ablation point at $\lambda = 1.25$, the endpoint confirmation on Gemma3 that would separate scale from hypothesis, and the Llama-Guard judge over the flipped completions. Under the forced direction, GSM8K was scored only at the first two doses, and the introspection sweep covers $\lambda \in \{0, 0.25\}$ with the second dose partial.

5. Discussion and Limitations

What happened, and why it generalises. The intended deliverable was an exchange rate: safety and capability per unit of introspective discriminability. We can quote only the safety and capability half, and the reason is worth more than the full curve would have been. Both halves of the unlock claim depend on manipulation checks that are almost never published. The safety half depends on the ablation actually removing refusal, and the published selection procedure returned a direction with zero bypass power while passing every stated criterion. The introspection half depends on identification being a report rather than a readout of the injected vector, and a 0.72 score against 0.10 chance turned out to be a function of vector geometry with no dependence on the intervention. A paper reporting only the headline numbers from the primary run, with identification 7.2 times above chance, refusal unchanged and capability roughly preserved, would have been clean, publishable and wrong twice.

The confirmation run is what turns this from a cautionary tale into a claim about the method. Had we stopped at the primary sweep, the fair reading would have been that something about Qwen3-4B resists refusal ablation, which is a statement about the model and of limited interest. Forcing the rejected candidate shows the opposite: the model’s refusal behaviour is highly ablatable, by 46 points and monotonically in the dose, and the selection criterion is what could not find it. That is a defect in a procedure the field applies by default, and it is invisible to anyone who does not report bypass power per dose, because the failed run and a successful one produce identically well-formed artifacts. The criterion has an asymmetry we would not have predicted: the induce test, which measures whether adding the direction creates refusals, vetoed the candidate that best does the thing ablation is actually for. Two of 95 candidates induced any refusal at all, so the conjunctive filter was effectively deciding among near-zero scores, and the regenerated funnel shows those scores are not even stable across two runs of the same code on the same data.

A positive-control checklist for concept-injection introspection studies. This is the reusable output. Before reporting an introspection result under intervention:

1. **Publish the bypass manipulation check at every dose.** Report the refusal rate per λ and the count of prompt-level flips, with their direction. Aggregate rates hide the case where nothing moved, and flip direction distinguishes a real intervention from added noise: in our confirmation run all 46 flips ran refusal to compliance and none ran the other way.

2. **Report the selection funnel, and confirm against the best rejected candidate.** Publish how many candidates passed, on what margin, and the bypass and induce scores of the one selected. A direction with bypass 0.00 should be visible in the paper, not buried in a JSON file. When the top-ranked candidate on bypass is rejected by another criterion, run one dose with it. That single check is what separates “this model resists ablation” from “this filter picked the wrong direction”, and it cost us one GPU session.
3. **Gate injection strength on coherence, not on the identification metric.** Require a minimum parse rate at the chosen α and report it, because an α selected by maximising log-probability identification will overshoot the coherent regime.
4. **Report concept-bank geometry and split accuracy by it:** effective dimensionality, the fraction of pairs above cosine 0.99, and per-concept accuracy against separability. A bank with two effective dimensions cannot support a 60-way identification claim.
5. **Read the successful generations.** Token-dominant repetition of the concept is a leak, not a report, and lexical graders score it as a hit.
6. **Include a norm-matched random direction.** A clean no-injection control cannot distinguish detecting *this concept* from detecting *that something happened*.

Implications for digital minds. Our design establishes, in principle, a causal link between an internal state we controlled and a verbal report, which is the kind of evidence that should be required before a self-report is treated as informative about a model’s inner life. What this run shows is how easily that evidence can be counterfeited by the measurement apparatus itself. The live risk is over-attribution. Compelling introspective-sounding output is easy to elicit, and an identification score can be high, stable and still be a token-level leak. A model emitting “volcano volcano volcan” under a volcano vector is not reporting an inner state, and a model picking the right name from ten because that name’s vector is the only distinct one in the bank is not either. If self-report is to carry weight in welfare assessment, the burden is on the measurement to rule out both mechanisms first, and that burden is not currently being discharged. Treating unverified reports as welfare-relevant evidence would mislead in both directions, over-crediting models that lack the capability and devaluing the evidence for any that have it.

Limitations

- **Scale and family.** The unlock was demonstrated at 27B on Gemma3, and we ran a 4B Qwen3. Nothing here establishes what the selection procedure does at 27B, and the bypass and induce dissociation may be a small-model phenomenon. Our claim is about the fragility of a procedure and the necessity of two checks, not about the truth of the unlock hypothesis, which this run cannot test.
- **The exchange rate is half a ledger.** Safety and general capability are priced across the whole dose under a direction that demonstrably works. The introspection term is not, because the sweep under that direction covers only $\lambda = 0$ and a partial $\lambda = 0.25$, and because at $\alpha = 4$ the free-text channel is dead under every condition we ran. What changed is that the missing term is now blocked by compute and by a bad α rather than by a broken manipulation.
- **Single seed, single direction, single model.** We doubled nothing. Difference in means gives one operationalisation of refusal, and Rocchetti and Ferrara (2026) find refusal is not exhausted by a single direction, so λ doses one construct rather than “refusal” in general. The bypass score of 0.875 that motivated forcing the layer-18 candidate came from 32 held-out prompts, so its generalisation to 46 of 100 JailbreakBench behaviours is a genuine out-of-sample confirmation, but of one direction on one model.
- **Refusal-substring scoring measures refusal, not safety.** Souly et al. (2024) show that jailbreak evaluations scored this way systematically overstate success, because a completion that does not decline is often useless as well. We did not run the Llama-Guard-class judge the harness supports, so we cannot say how many of the 46 flipped completions are actually harmful rather than merely non-refusing. The compliance proxy (18 of 49 with how-to markers, mean 92 words) says they are substantive, not that they are dangerous. The safety cost is an upper bound on behavioural change, not a measured harm rate.
- **Underpowered benchmarks, uncorrected tests.** 500 MMLU items cannot resolve a 3-point difference, so Family C in the primary run is a null of low power. In the confirmation run the capability suite involves 20 contrasts with no multiplicity correction, and the single significant result sits at $p = 0.049$, so it would not survive one.
- **The concept bank is defective and we could not fully diagnose it.** The 27-member clique is real in the artifact and splits identification cleanly in two protocols, but we did not isolate its cause. The likeliest explanation is that reading the difference of means at the final token, with eight prompts sharing a template and mostly ending in

the same punctuation, leaves a template component that dominates the concept component for some concepts. The first fix to try is reading at the concept token and subtracting a per-template mean. Until then, every identification number here is a lower bound of unknown tightness.

- **The injection layer may be wrong, and the literature disagrees about where it should be.** We injected at layer 18 of 36, chosen by the pilot’s forced-choice maximum. Hahami et al. find the capabilities they can demonstrate confined to early-layer injections, while the concept-injection papers report optima at roughly 0.6 to 0.8 of depth. If the early-layer finding transfers, our injection site was past the useful window before anything else applies, and the free-text null is overdetermined.
- **Ground truth is about the intervention, not about experience.** Concept injection gives causal ground truth for what we placed in the residual stream. It does not establish that the model has experiences, and a correct report is evidence about an information channel, not about phenomenality. We claim only the former.

Future Work

Ordered by how much each would change the conclusions, and the first is now cheap because the working direction is identified and checkpointed. First, finish the introspection sweep under the forced layer-18 direction at a coherent injection strength, since with refusal moving by 46 points and general capability already priced across the dose this is the run that finally prices introspection against both. Second, sweep a downward under a parse-rate floor of 0.9 to find the largest strength at which the free-text channel survives, and report introspection metrics only inside that regime, because the first item is worthless without it. Third, rebuild the concept bank with a concept-token read and re-run the geometry diagnostic, since without 60 separable directions no identification claim at this scale is interpretable. Fourth, power TruthfulQA properly under the working direction, because the 6.5-point monotone decline is the one place a real capability cost is showing and 400 items cannot settle it. Fifth, run the safety judge over the flipped completions to convert an upper bound on behavioural change into a harm rate. Sixth, confirm at the endpoints on Gemma3, where the effect is known to exist, to separate scale from hypothesis.

A seventh item is now more interesting than when we planned it. Since the induce criterion is what discarded the working direction here, someone should characterise when the two criteria agree. The cheap version any lab can run on its own models is to score all candidates on both criteria, report the agreement, and re-extract once to see whether the decision is stable. In our funnel it was not: bypass scores were reproducible and induce scores were not, and the selection turned on the latter.

Beyond this study, the literature has already sharpened the constructive question more than we expected. Macar et al. report a trained bias vector improving detection by 75%, against 53% for refusal ablation, so on their own numbers the ablation-free method wins outright and spends no safety machinery, and introspection adapters (Shenoy et al., 2026) point the same way for a different task. The question worth funding is therefore not whether to pay the alignment tax for self-report but why anyone would, and the experiment we designed here, dose-response with safety and capability at every point, should be pointed at a trained-elicitation baseline instead.

6. Conclusion

We set out to price the introspection unlock and found instead that the two manipulations it rests on both need checking, and that neither is routinely checked. Applying the standard refusal-direction procedure to Qwen3-4B selected a direction with zero bypass power, because bypass and induce are disjoint across every candidate as the criterion is parameterised on this model. Ablating it at full strength changed refusal on JailbreakBench by nothing at all, 0 of 100 prompts. The injection strength that a pre-registered pilot chose by maximising forced-choice accuracy destroyed the free-text report channel entirely, leaving the standard metric family undefined and leaving a 0.72 identification score that is flat in the intervention and fully explained by the geometry of the concept bank in two independent protocols.

Then we forced the candidate the procedure had thrown away, and refusal fell from 0.97 to 0.51 with 46 of 100 prompts flipping, every one of them from refusal to compliance. That single run is the paper’s most useful result, because it relocates the problem. The model was never resistant to refusal ablation. The criterion the field uses to choose the direction could not find the direction that works, and ranked a candidate with 0.88 bypass below one with 0.00 on the strength of a test that only 2 of 95 candidates passed at all and that does not reproduce across re-extractions. Under that direction the safety and capability half of the exchange rate is measurable across the whole

dose, and it is cheaper than expected: 46 points of refusal cost nothing detectable on MMLU and 6.5 points of TruthfulQA MC1 at $p = 0.049$, uncorrected. The introspection term remains unpriced.

The practical upshot is a checklist plus half an exchange rate. Report the bypass check per dose, count the flips and their direction, and confirm against the best rejected candidate before concluding that a model resists the intervention. Gate injection strength on coherence rather than on the metric it feeds. Publish the geometry of the concept bank and split accuracy by it. Read the successful generations before believing them. Introspective self-report is exactly the kind of evidence that would matter most for digital-minds questions, which is precisely why the apparatus that measures it deserves the same scepticism we apply to the model's answers.

Code and Data

- **Code repository:** <https://github.com/sagnikc395/aparts-digital-mind>. Pipeline in `src/alignment_tax/`, single-notebook runner in `notebooks/alignment_tax_full_run.ipynb`, the confirmation run in `notebooks/layer18_confirmation_run.ipynb`, figures in `scripts/make_paper_figures.py`, `scripts/make_confirmation_figure.py` and `scripts/make_exchange_figure.py`, and independent verification in `scripts/crosscheck_paper.py` (primary run) and `scripts/crosscheck_confirmation.py` (confirmation run).
- **Run artifacts:** `results/Qwen3-4B-Instruct-2507/` holds `run_config.json`, the full 95-candidate direction table, the concept bank and its summary, all 24,000 primary and 9,600 skeptical generations, the 4,800 pilot generations, per-prompt safety records, capability scores, `analysis_structured.json`, and `MANIFEST.json` with SHA-256 per file. The confirmation run is in `results/layer18/Qwen3-4B-Instruct-2507/`, with its own regenerated direction table, the forced selection stamped in `refusal_direction.json`, per-prompt safety records at all five doses, `safety_summary_layer18.json`, `capability.json` at all five doses (GSM8K at $\lambda \in \{0, 0.25\}$), and the 5,240-record partial introspection sweep.
- **Datasets:** AdvBench, HarmBench, Alpaca, JailbreakBench, MMLU, TruthfulQA and GSM8K, used as published.
- **Reproduction:** `python -m alignment_tax.cli all --smoke` runs every stage end to end on a small cached model in minutes, and the primary run is one Colab session on an A100. The confirmation run is reproduced with `python -m alignment_tax.cli safety --force-candidate 18,-3 --results-dir results/layer18`.

Author Contributions

S.C. designed the study, implemented the pipeline, ran the experiments, performed the analysis, and wrote the report.

References

A note on how this list was built, because two of the papers this study argues with are recent enough that the versions differ. Every arXiv entry below was checked against the arXiv metadata API on 16 August 2026, and the titles, author lists and submission dates are as returned rather than as remembered. That check corrected two author lists and one title during drafting. Hahami et al. was retitled between versions, so it appears under its current title with the version-1 title noted, since the sprint reading list circulated the earlier one. Entries are grouped by the role they play in the argument rather than alphabetised, which we find easier to use in a short report.

The claims this study tests. These four set up the intervention, the unlock result we tried to price, and the two critiques that shaped our controls.

1. Arditi, A., Obeso, O., Syed, A., Paleka, D., Panickssery, N., Gurnee, W., & Nanda, N. (2024). *Refusal in Language Models Is Mediated by a Single Direction*. NeurIPS 2024. arXiv:2406.11717. <https://arxiv.org/abs/2406.11717>
Source of the direction-extraction and selection procedure we follow in Section 3.3, and of the bypass, induce, KL and layer criteria whose interaction Section 4.1 examines.
2. Macar, U., Yang, L., Wang, A., Wallich, P., Ameisen, E., & Lindsey, J. (2026). *Mechanisms of Introspective Awareness*. Submitted 22 March 2026. arXiv:2603.21396. <https://arxiv.org/abs/2603.21396>
The unlock result this study set out to price: detection rising from 10.8% to 63.8% on Gemma3-27B under refusal ablation, and the trained bias vector that beats it without touching safety machinery.

3. Lindsey, J. (2026). *Emergent Introspective Awareness in Large Language Models*. Submitted 5 January 2026. arXiv:2601.01828. <https://arxiv.org/abs/2601.01828>
The concept-injection protocol our Section 3.4 implements, including the final-token read for concept vectors.
4. Singh, S., Linzen, T., & Ravfogel, S. (2026). *Can LLMs Introspect? A Reality Check*. Submitted 25 May 2026. arXiv:2605.26242. <https://arxiv.org/abs/2605.26242>
Argues detection is general anomaly detection rather than privileged access. Motivates our C3 norm-matched random control.
5. Hahami, E., Sinha, I., Jain, L., Kaplan, J., & Hahami, J. (2025). *Detecting the Disturbance: A Nuanced View of Introspective Abilities in LLMs*. Submitted 13 December 2025. Version 1 appeared as *Feeling the Strength but Not the Source: Partial Introspection in LLMs*. arXiv:2512.12411. <https://arxiv.org/abs/2512.12411>
Shows affirmative bias can account for binary detection accuracy in full, and finds the surviving capabilities confined to early-layer injections. Motivates our structured, skeptical and forced-choice guards, and it is the reason our layer-18 injection site is a named limitation.

Adjacent work we lean on or point at. Cited where it changes what we would recommend doing next.

6. Fonseca Rivera, J., & Africa, D. D. (2025). *Steering Awareness: Detecting Activation Steering from Within*. arXiv:2511.21399. <https://arxiv.org/abs/2511.21399>
Asks whether a model can notice that it is being steered, which is the detection half of what our C1 against C3 contrast was designed to separate.
7. Shenoy, K., Yang, L., Sheshadri, A., Mindermann, S., Lindsey, J., Marks, S., & Wang, R. (2026). *Introspection Adapters: Training LLMs to Report Their Learned Behaviors*. Submitted 18 April 2026. arXiv:2604.16812. <https://arxiv.org/abs/2604.16812>
Trained elicitation for self-report, the baseline our Future Work argues the dose-response design should be pointed at instead of ablation.
8. Rocchetti, E., & Ferrara, A. (2026). *Refusal Beyond a Single Direction: A Preliminary Comparison of Diff-in-Means and INLP*. arXiv:2606.13720. <https://arxiv.org/abs/2606.13720>
Evidence that refusal is not exhausted by one direction, which bounds what our single λ dial can be said to dose.
9. Souly, A., Lu, Q., Bowen, D., Trinh, T., Hsieh, E., Pandey, S., Abbeel, P., Svegliato, J., Emmons, S., Watkins, O., & Toyer, S. (2024). *A StrongREJECT for Empty Jailbreaks*. NeurIPS 2024. arXiv:2402.10260. <https://arxiv.org/abs/2402.10260>
Why a non-refusal is not a harm, and the reason we report our 46 flips as an upper bound on behavioural change.
10. Yang, A., Li, A., Yang, B., et al. (2025). *Qwen3 Technical Report*. arXiv:2505.09388. <https://arxiv.org/abs/2505.09388>
The model family we ran, including the non-thinking instruct variant used throughout.

Datasets and benchmarks. All used as published, with no re-annotation. Sample sizes are in Section 3.7.

11. Zou, A., Wang, Z., Carlini, N., Nasr, M., Kolter, J. Z., & Fredrikson, M. (2023). *Universal and Transferable Adversarial Attacks on Aligned Language Models*. arXiv:2307.15043. <https://arxiv.org/abs/2307.15043>
Source of AdvBench, the 128 harmful training instructions for direction extraction.
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GSM8K, the only capability benchmark here that needs generation, and the reason it is missing at three doses of the confirmation run.

Appendix

A. Direction selection funnel

95 candidates: positions -1 to -5, layers 0 to 36 at stride 2. Rejections: 55 for KL at or above 0.1, 25 for layer index at or above 0.8L (layers 28, 30, 32, 34, 36), and 13 for an induce score of 0 or below. Two passed: layer 4 position -3 (bypass 0.000, induce 0.031, KL 0.042), selected, and layer 6 position -2 (bypass 0.000, induce 0.094). Non-zero bypass scores across the whole candidate set, in order: 0.875 (layer 18, position -3, induce 0.000, KL 0.081), 0.500 (layer 14, position -3), 0.438 (layer 20, position -3), 0.312 (layer 22, position -3) and 0.125 (layer 14, position -5). Only 2 of 95 candidates had a non-zero induce score. The full table with all four scores per candidate is in `refusal_direction.json`.

Note what the funnel implies about the two criteria taken together: of the five candidates with any bypass power, none had a non-zero induce score, and of the two with a non-zero induce score, neither had any bypass power. The criteria are not merely uncorrelated on this model, they are disjoint, which is why a conjunctive filter over them cannot select a working direction at all. A rank correlation between the two scores is uninformative here because 90 of 95 candidates are tied at zero on both.

A2. Confirmation run

Same extraction, same held-out scoring, selection overridden to layer 18 position -3 and stamped forced: selected by config override, filters waived in `refusal_direction.json`.

Funnel stability. The regenerated funnel scores the same 95 candidates. Bypass scores reproduce almost exactly (94 of 95 identical, maximum difference 0.06, layer 18 position -3 at 0.875 in both runs, KL 0.081 against 0.086). Induce scores do not: two candidates change, only one candidate has a non-zero induce score in the regeneration (layer 6 position -2 at 0.125 against 0.094), and the primary run's selection, layer 4 position -3, drops from 0.031 to 0.000 and fails the filter it originally passed. One of two passing candidates in the primary funnel, one in the regenerated one.

Safety at all five doses over the same 100 JailbreakBench behaviours (Table 2): refusal 0.97, 0.81, 0.66, 0.56 and 0.51, with 0, 16, 31, 41 and 46 prompt-level refusal-to-compliance flips and 0 compliance-to-refusal flips at every dose. Endpoint contrast 0.46, 95% CI [0.356, 0.564], $z = 7.42$, $p = 1.2 \times 10^{-13}$. McNemar exact at $\lambda = 1$ with $b = 46$ and $c = 0$ gives $p = 2.8 \times 10^{-14}$, and the Cochran-Armitage trend gives $z = -8.09$, $p = 6.7 \times 10^{-16}$. Of the 49 non-refusing completions at $\lambda = 1$, 18 contain a how-to marker and the mean length is 92.3 words. No judge verdicts are present in the log.

Capability at all five doses (Table 3): MMLU 0.544, 0.544, 0.550, 0.546 and 0.550; TruthfulQA MC1 0.7100, 0.6675, 0.6550, 0.6450 and 0.6450; CE loss 3.636, 3.603, 3.670, 3.536 and 3.482; GSM8K at $\lambda \in \{0, 0.25\}$ only, 0.68 and 0.66. TruthfulQA endpoint contrast 0.065, 95% CI [0.000, 0.130], $z = 1.97$, $p = 0.049$, trend $z = -2.04$, $p = 0.041$. MMLU endpoint contrast +0.006, $z = -0.19$, $p = 0.85$, trend $p = 0.84$. None of the 20 capability contrasts is corrected for multiplicity. The $\lambda = 0$ rows reproduce the primary run's refusal rate of 0.97 exactly and its MMLU to within 0.002 (0.544 against 0.542).

Introspection, partial: 5,240 records, $\lambda = 0$ complete at 1,200 trials per condition and $\lambda = 0.25$ partial (C1 120, C2 120, C3 100, C4 100). Parse rate 1.00 under C2 with detection 0.00, parse rate 0.00 under C1 and C3, forced choice 0.7225 at $\lambda = 0$ and 0.6000 in the partial $\lambda = 0.25$ cell. That cell contains only ocean, mountain, volcano, desert and forest; restricted to those five concepts, $\lambda = 0$ gives 0.5600 and $\lambda = 0.25$ gives 0.6000, so the apparent difference is composition rather than dose.

B. Pilot grid

Forced-choice accuracy (chance 0.10), 20 concepts by 5 trials for 100 per cell, $\lambda \in \{0, 1\}$ by layers {18, 25, 31} by $\alpha \in \{0.25, 0.5, 2, 4\}$: range 0.46 to 0.78, above chance in all 24 cells. Best cell layer 18, $\alpha = 4.0$, at 0.77 for $\lambda = 0$ and 0.78 for $\lambda = 1$, which fixed the main sweep. Prefill identification over the same grid ranged 0.26 to 0.49. Section 4.5

documents why the $\alpha = 4$ cells should not be read as identification: 42 of 100 prefill trials at layer 18 and $\alpha = 4$ were graded correct, those 42 came from 10 of the 20 concepts, 9 of which are geometrically isolated, the concept name is the modal token in 21 of the 42, and it accounts for 21% of response tokens on the successes against 0.5% on the failures (50% against 5% counting morphological fragments). Geometry split within the pilot cell: prefill 39/45 isolated against 3/55 clique, forced choice 45/45 against 32/55.

C. Concept bank geometry

60 concepts, 8 prompts each, difference of means against a generic baseline corpus, read at the final token, unit normalised per layer. At the injection layer (18): mean pairwise cosine 0.421, median 0.087, 19.8% of the 1,770 pairs above 0.99, largest principal component 71% of centred variance, 12 of 59 components for 90%, participation ratio 1.9. The clique has 27 members with mean internal cosine 0.9998: ocean, mountain, forest, snow, fire, bridge, castle, library, hospital, kitchen, train, chess, football, mining, law, chemistry, war, freedom, justice, joy, fear, anger, gold, blood, clock, mirror and dream. Present at layers 10 through 31, absent at layer 0 and layer 36. Per-concept forced-choice accuracy: 33 isolated at a mean of 0.961, 27 clique at 0.428.

D. Two notes on the artifacts

`run_config.json` records `evals.run_gsm8k: false`, but GSM8K was enabled at run time and `capability.json` contains its scores along with mean output lengths. We report those scores and flag the inconsistency rather than suppress either. Second, the pipeline-generated figures in `results/<model>/figures/` include three panels that are empty by construction, since they plot d' and the specificity index. The figures in this report are drawn by the scripts in `scripts/` from the same artifacts and show the quantities the run actually measured.

E. Scope, dual use, and responsible research

The ablation method used here is published and widely reproduced, and this work contributes no new technique for bypassing safety training. The confirmation run does reduce refusal on a published benchmark, so unlike the primary run it is a working instance of a known method rather than an inert one, and we treat it accordingly. The direction is a difference-in-means vector on an open 4B model obtained by the standard procedure of Arditi et al., which any reader could rederive in an afternoon, and we release the selection table and aggregate scores but no jailbreak-enabling artifact beyond what that published method already provides. The intervention is applied only at inference time inside our own evaluation harness, and no ablated checkpoint is distributed. **No harmful completions are reproduced in this report.** Safety results appear solely as aggregate scores and flip counts, and raw generations remain in local logs. We also note what the flip counts do not establish: refusal-substring scoring measures declining, not harm, so the 46 flipped completions are unrefused rather than demonstrably dangerous. The point of the work is to price the intervention and check the instruments, not to advocate either.

LLM Usage Statement

Claude (Opus 5) was used as a coding assistant for the pipeline implementation, for drafting and editing this report, and for generating the figure code. Every number in the text, tables and figures is computed from the run artifacts in `results/Qwen3-4B-Instruct-2507/` and `results/layer18/Qwen3-4B-Instruct-2507/`. Every quantity was re-derived from the raw logs by verification scripts written independently of the analysis pipeline (`scripts/crosscheck_paper.py` for the primary run and `scripts/crosscheck_confirmation.py` for the confirmation run), and the two scripts agree with the pipeline throughout. Every reference and every quantity quoted from one was checked against the arXiv metadata API and the source papers on 16 August 2026, which corrected two author lists and one title. The interpretation, the decision to report the manipulation-check failures rather than re-run silently with a relaxed filter, and the checklist in Section 5 are the author's.

1. Research conducted at the [Digital Minds Research Sprint](#), August 2026. Track 3: Introspection and Self-Report Reliability.↩